

1 Question (15 points)

Write a recursive method

```
int aBeforeB(String s)
```

which returns the number of 'a' characters preceding a 'b' character in the string *s*.

2 Question (20 points)

Write a method that takes a matrix as parameter (2 dimensional array) and returns true if the matrix is a row increasing matrix, false otherwise. A matrix is row increasing if the numbers in each row are in increasing order.

3 Question (15 points)

Write a recursive method

```
int S(int N, int k)
```

to calculate Stirling numbers of the first kind where

$$S(n, n) = 1$$

$$S(n, 0) = 0, \text{ if } n \neq 0$$

$$S(n, 0) = 1, \text{ if } n = 0$$

$$S(n, k) = kS(n-1, k) + S(n-1, k-1)$$

4 Question (20 points)

Write a program that prompts the user to enter the number of websites, the domain names of each websites (e.g. boredpanda.com, onedio.com, sosyalon.com), and the number of visitors of these websites (e.g. 2255, 3578, 1185). Then, the program prints the list into an output file that has the websites in *decreasing order of their number of visitors*.

```
Enter number of websites: 3
```

```
Enter the name of website: boredpanda.com
```

```
Enter the number of visitors: 2255
```

```
Enter the name of website: onedio.com
```

```
Enter the number of visitors: 3578
```

```
Enter the name of website: sosyalon.com
```

```
Enter the number of visitors: 1185
```

```
Top Websites:
```

```
1) onedio.com 3578
```

```
2) boredpanda.com 2255
```

```
3) sosyalon.com 1185
```

In this program, you have implement sub-operations as *methods*. You are expected to write ***at least 2 methods***.

5 Question (15 points)

Write a function **insert**, which takes a sorted array A and a value k as arguments, and returns a new array which contains all values of array A and the value k in a sorted manner. You are not allowed to use any sort function in this problem.

$A = \{2, 4, 5, 6\}$, $k = 3$.

$\{2, 3, 4, 5, 6\}$

$A = \{3, 4, 8\}$, $k = 10$.

$\{3, 4, 8, 10\}$

6 Question (15 points)

Suppose that a spaceship game area is modeled by a two dimensional 100 by 100 integer array. In this matrix, enemies are represented by value 1 and empty cells are represented by value 0. The spaceship can be at any position and hit the enemies to destroy them. The direction of the spaceship is always up (north) and the spaceship can hit an enemy if both of them are on the same column. If there are more than one enemies in front of the spaceship on the column, only the closest enemy will be destroyed. If an enemy is hit, its corresponding value at the array must be turned into 0.

Write a method named **hitEnemy**, which takes a two dimensional array (game area) and the position of the spaceship with its column and row indices (cx,cy) as arguments to simulate spaceship fire. An example matrix representing a 10x10 game area is shown below.

	1		1		1		1		1		1		1		1		1		1
		1				1													
						1													

1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	0	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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The **hitEnemy** method checks whether the fire from position cx-cy hits an enemy or not. When it hits the first enemy, it removes the enemy (turns 1 into 0) from the game matrix and returns true. If there is no enemy to hit on that column, it returns false.